

Total No. of Questions : 8]

SEAT No. :

PE4260

[6582] 32

[Total No. of Pages : 2

S.E. (Electronics and Computer Engineering)

ELECTRONIC CIRCUITS

(2019 Pattern) (Semester - III) (204202)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Draw and explain the block diagram of LM317. [6]
b) Define the line and load regulation in case of voltage regulator. What are the ideal values for same? [6]
c) Write the short note on low drop out voltage regulator. [6]

OR

- Q2)** a) Explain current boosting voltage regulator using LM317. [6]
b) Determine the range over which voltage range can varied in LM317 voltage regulator if the values of R_1 is 240 ohm and R_2 taken as 5K potentiometer. Assume $I_{adj} = 100\mu A$. Draw the typical connection diagram. [6]
c) Compare linear and switching power supply. [6]

- Q3)** a) Draw internal the block diagram of OPAMP and explain in detail. [6]
b) Explain the following op-amp parameters: [6]
i) Input offset voltage
ii) CMRR
iii) Slew Rate
c) What is current mirror? Why it is used? [5]

OR

- Q4)** a) Explain the following parameter of Op-Amp [6]
i) Slew rate
ii) PSRR
iii) Gain bandwidth product
b) Explain the level shifter stage of Op-Amp in detail. [5]
c) Justify, how constant current source used in placed of R_E to improve the CMRR in differential Amplifier. [6]

P.T.O.

- Q5) a)** Draw neat circuit diagram of: [6]
i) Schmitt trigger
ii) Precision amplifier
b) Explain Zero crossing Detectors using op-amp with necessary waveforms. [6]
c) List important characteristics of comparator. What are the advantages of Schmitt trigger over comparator? [6]

OR

- Q6) a)** Draw circuit diagram of three op-amp instrumentation amplifier and write its output equation. [6]
b) Explain Practical differentiator circuit with neat circuit diagram. What are the limitations of ideal differentiator? [6]
c) Draw the circuit diagram of practical differentiator and write equation for output voltage (V_o). [6]
- Q7) a)** Define the following terms w.r.t PLL: [6]
i) Free running Frequency
ii) Lock range
iii) Capture range
iv) Pull in Time
b) Draw the block schematic of PLL and explain each block in detail. [5]
c) List the various methods of A/D Conversion. Compare them and explain which one is fastest. [6]

OR

- Q8) a)** Draw block diagram and explain any one application of IC PLL 565 in detail. [6]
b) Write short notes on application of PLL : [6]
i) Frequency Multiplier
ii) FM Demodulator
c) Draw and explain successive approximation A/D converter. [5]

